A quadratic family of integral two-torsion classes in a Huneke–Wiegand Koszul presentation

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Scope. A quadratically growing direct summand of integral two-torsion, not a description of the whole connecting quotient.

Abstract

We study an explicit integral Koszul presentation arising in the graded-syzygy analysis of a previously constructed Huneke–Wiegand family. For every integer $p \geq 8$, its cokernel contains a direct summand isomorphic to $(\mathbb{Z}/2)^{\lfloor((p-2)^2+3)/12\rfloor}$. The classes are indexed by all strictly increasing nonnegative triples of sum $p-2$. Reflection-symmetric interval potentials give explicit original integral sources for twice each class. Relative parity functionals, supported on ten to twelve original rows, detect the classes independently and vanish on every complete connecting image; extension over the remaining target gives a nonconstructive retraction. A previously tracked four-row vector is integrally congruent to one selected class, with an explicit source for the difference. We also give its full twice-source, two short kernel-domain corrections, and a neighboring integral potential basis of rank $\binom{p-1}{2}$ with at most seven-row images. Independent finite campaigns check the signed identities and complete-sector parity certificates, preserving all tested sources in lossless archives. A failed local search is retained as an exact locality obstruction. The uniform theorem follows from signed reconstruction, exhaustive source classification, and an elementary counting recurrence, not finite rank extrapolation. The full cokernel, an upper bound, and identification with earlier isolated subpresentations remain unresolved.

1 Introduction and scope

Exact linear algebra can certify a finite presentation without revealing why the same phenomenon should persist with a parameter. This distinction matters for the connecting maps studied here: a displayed target vector, an integral multiple in the image, and a nonzero quotient class are different assertions. The source identity and relative parity argument close the second and third of these obligations for a quadratically growing independent family of target classes. Their subgroup is a direct summand, although its complement remains undetermined.

The underlying family and its conductor-fiber-cone and graded-syzygy context were developed in the first paper of this programme [1]. We make the present matrix problem self-contained by defining its complete integral presentation below. The new result is therefore checkable without reproducing the semigroup construction. Its application to that family uses the presentation identified in the earlier record; it does not repeat or claim priority for the discovery of a counterexample addressed by that programme.

Signed Koszul strands admit useful combinatorial descriptions. For semigroup quotients, Bruns and Herzog provide a relative simplicial viewpoint [2]. Here the arguments use explicit oriented bases and low-exterior complements. We do not infer a simplicial or projective-plane model from a matrix with entries $0, 1, -1$, and no unidentified topological model is a premise of the proof. Potentials and signed complements are classical tools; the contribution is the explicit triangle family in this presentation, including its full sources, all boundary corrections, and complete relative functionals. The general fact that vector-space functionals extend is used only to deduce splitting; it is not claimed as a new method.

Statements proved deductively are marked [D]; finite exact checks are marked [MV]. Remaining research questions are marked [C]. The annihilator theorem alone says *order dividing two*. The additional nonvanishing proof in Section 7 upgrades the tracked class to order exactly two. Section 8 constructs all triangle classes, proves their exact relation lattice and count, and derives the direct-summand corollary. These are lower-bound and partial-decomposition results, not the whole quotient.

Independent validation boundary. The generic parity, independence and splitting arguments below are deductive and have undergone independent paper review. The separate full-sector audit for all 27 triangles at five declared parameters also passes. Section 9 distinguishes these finite certificates from the exhaustive all-parameter proof; no finite rank pattern is extrapolated.

2 The complete integral presentation

All intervals in this paper contain integers, and an empty interval contributes nothing. Fix $p \geq 8$. Put

$$L_0 = [1, p], \qquad L_1 = [3p, 4p-2], \qquad L = L_0 \cup L_1, \qquad N = |L| = 2p - 1.$$

The high set is the following disjoint union:

$$\begin{aligned}
&H_0 = [6p, 8p-2], && H_1 = [8p, 10p-2], && H_2 = \{10p\},\\
&H_3 = [11p-1, 12p-1], && H_4 = [13p+1, 14p-2], && \\
&H_5 = [14p, 15p-1], && H_6 = \{16p\}, && H_7 = [17p-1, 18p-1].
\end{aligned}$$

Write $H = \bigcup_{i=0}^{7} H_i$ and $G = L \cup H$. The degree-two high coefficient set is $\mathcal{B} = [6p, 24p-1] \setminus H$, explicitly

$$\begin{aligned}
\mathcal{B} = \{8p-1, 10p-1, 14p-1\} \cup [10p+1, 11p-2] \cup [12p, 13p]\\
\cup [15p, 16p-1] \cup [16p+1, 17p-2] \cup [18p, 24p-1].
\end{aligned}$$

For later labels, $\mathrm{C}_0 = \{8p-1\}$ and $\mathrm{C}_2 = [10p+1, 11p-2]$. These are names for coefficient subsets, not subquotients asserted to split off.

Define a low product with values in formal degree-two symbols by

$$\mu(a,c) = \begin{cases} \mathsf{A}_{a+c}, & a, c \in L_0,\ a+c > p, \\ \mathsf{B}_{a+c}, & a, c \text{ lie in different } L_i,\ a+c \geq 4p-1, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

The total offset of the strand is

$$\tau = 4p^2 + 6p - 1 = \sum_{a \in L} a + 10p - 2, \qquad \sum_{a \in L} a = 4p^2 - 4p + 1.$$

Let $\mathcal{S}_p$ be free over $\mathbb{Z}$ on all labels $S[E;c]$ with $E \subseteq G$, $|E| = 2p-2$, $c \in L$, and $\sum E + c = \tau$. Let $\mathcal{K}_p$ have the analogous basis $K[E;c]$ with $c \in H$. All exterior sets are ordered increasingly. The target is free on labels

$$D[E;\mathsf{A}_d], \quad D[E;\mathsf{B}_d], \quad K[E;b], \qquad |E| = 2p-3, \quad \sum E + d = \sum E + b = \tau,$$

where $d \in [p+1, 2p]$ for type A, $d \in [4p-1, 5p-2]$ for type B, and $b \in \mathcal{B}$. Only the appropriate offset is used in each label. Denote the D and K target summands by $\mathcal{D}'_p$ and $\mathcal{K}'_p$.

For $E = \{e_0 < \cdots < e_{2p-3}\}$, define

$$\begin{aligned} M_p S[E;c] &= \sum_{\substack{i: e_i \in L \\ \mu(e_i,c) \neq 0}} (-1)^i D[E \setminus \{e_i\}; \mu(e_i,c)] \\ &\quad + \sum_{\substack{i: e_i \in H \\ e_i + c \in \mathcal{B}}} (-1)^i K[E \setminus \{e_i\}; e_i + c], \end{aligned} \tag{2}$$

$$M_p K[E;c] = \sum_{i: e_i + c \in \mathcal{B}} (-1)^i K[E \setminus \{e_i\}; e_i + c]. \tag{3}$$

Thus $M_p : \mathcal{S}_p \oplus \mathcal{K}_p \longrightarrow \mathcal{D}'_p \oplus \mathcal{K}'_p$ is completely specified over the integers. There is no implicit projection in these definitions. Our quotient is $\operatorname{coker}_{\mathbb{Z}} M_p$, and a source identity always refers to this entire map.

3 The four-row target and its earlier representative

Set

$$G_p(a,j;c) = K[(L \setminus \{a, 3p, 3p+j\}) \cup \{6p\}; 10p + c].$$

The tracked endpoint vector is

$$\begin{aligned} \eta_p = (-1)^p \big(& 2G_p(p-3, 2; p-3) - G_p(p-2, 2; p-2) \\ & + 2G_p(p-2, 1; p-3) - 2G_p(p-3, 1; p-4) \big). \end{aligned} \tag{4}$$

These are four distinct original rows. Their coefficient offsets are $11p-3, 11p-2, 11p-3, 11p-4$, all in C_2. Exactly one row coefficient is odd. This is a statement about a vector, not a nonvanishing result in the cokernel.

For completeness, define the larger D representative without a choice of quotient coordinates. Put

$$T(a,j) = S[(L \setminus \{a, 3p, 3p+j\}) \cup \{6p, 10p\}; a + j - 2]$$

and set $s_p = \sum w_j(a) T(a,j)$, with the only nonzero weights

j	a	$w_j(a)$
2	$1 \le a \le p-4$	$(-1)^{a+1}$
2	$a = p-3$	$(-1)^{p+1}$
1	$a = p-2$	$2(-1)^{p+1}$
1	$a = p-3$	$2(-1)^{p}$

Let b_p^A and b_p^B be respectively the type-A and type-B parts of $M_p s_p$, and write $b_p = b_p^A + b_p^B$. This defines them explicitly through (1)–(2); it is the representative recorded in EXP-052 and derived uniformly in EXP-056. Define also

$$q_p = K[(L \setminus \{3p, 3p+2\}) \cup \{6p\}; 10p].$$

Proposition 3.1 ([D). *Original representative transfer] For every $p \ge 8$,*

$$M_p(s_p + q_p) = b_p + \eta_p. \tag{5}$$

Consequently $[b_p] = -[\eta_p]$ in $\operatorname{coker}_{\mathbb{Z}} M_p$.

Proof. Every coefficient of s_p lies in $[1, p-3]$. Its $6p$ face vanishes, while its $10p$ face has negative sign and survives. Hence $M_p s_p = b_p + \gamma_p$ with $\gamma_p = -\sum w_j(a) G_p(a, j; a+j-2)$. In q_p, the first-low faces $a = 1, \dots, p-2$ survive with sign $(-1)^{a-1}$, and the other first-low faces vanish in H_3. Every present second-low face vanishes in H_4, and the $6p$ face vanishes at coefficient offset $16p$. Thus

$$M_p q_p = \sum_{a=1}^{p-2} (-1)^{a-1} G_p(a, 2; a).$$

Adding this expression to γ_p cancels the interval $1 \le a \le p-4$ and leaves exactly (4). □

4 A complete integral potential basis in one sector

First fix the exterior high set $\{6p, 8p-4\}$, not the high set used later for the final annihilator. Index first-low variables as $p - r$, $0 \le r \le p-1$, and second-low variables as $3p+u$, $0 \le u \le p-2$. There are exactly two possible source families:

$$A_u(r, s) = S[(L \setminus \{p-r, p-s, 3p+u\}) \cup \{6p, 8p-4\}; p+2+u-r-s],$$
$$B_r(u, v) = S[(L \setminus \{p-r, 3p+u, 3p+v\}) \cup \{6p, 8p-4\}; 3p+2+u+v-r].$$

Here $r < s$, $u+2 \le r+s \le p+u+1$ in the first family. In the second, $u < v$ and

$$\max(0, u+v-p+4) \le r \le \min(p-1, u+v+2).$$

Indeed, the coefficient offset is the sum of the three omitted low variables minus $4p-2$. Three omitted first-low variables make it negative; three omitted second-low variables make it exceed $4p-2$. The remaining two types give exactly these bounds. Thus no original source in this fixed-high sector has been omitted.

Theorem 4.1 ([D). *Integral potential basis] Let d_D be the D component of M_p restricted to this sector. Choose arbitrary integral potentials $f_u(r)$ vanishing for $r \le u+1$. For $r < s$ put $\alpha_u(r, s) = f_u(s) - f_u(r)$. For $u < v$ put $R = u+v$, $F = f_u - f_v$, and*

$$\beta_r(u, v) = \begin{cases} F(r) - F(R+3), & R \le p-4, \\ F(r), & R \ge p-3. \end{cases}$$

The source with actual coefficients

$$(-1)^{p+r+s+u}\alpha_u(r,s) \quad \text{on } A_u(r,s), \qquad (-1)^{p+r+u+v}\beta_r(u,v) \quad \text{on } B_r(u,v)$$

parametrizes $\ker_{\mathbb{Z}} d_D$ *uniquely. Its unit potentials, indexed by* $0 \le u \le p-3$*,* $u+2 \le r \le p-1$*, form an integral basis of rank* $\binom{p-1}{2}$*. Each basis source has coefficient height one and at most* $3p-5$ *terms; its full* M_p *boundary has at most seven K rows and no D rows.*

Proof. We give the signed reconstruction, including truncation boundaries. Complement an increasing low exterior E inside L, and orient its omitted variables in increasing first-endpoint order followed by increasing second-endpoint order. For missing positions $d_1, \dots, d_k$ in L, the unreordered complement sign is

$$(-1)^{k(N-1)-\sum d_i - k(k-1)/2}.$$

For $k=3$, reversing the omitted first-low variables gives exactly the two source signs in the theorem. Complementing contraction gives wedge insertion, with common factor $(-1)^{|E|-1} = -1$ here. Therefore all nonzero D equations, up to a common unit sign on each row, are

$$\alpha_u(r,s) - \alpha_u(r,t) + \alpha_u(s,t) = 0, \tag{6}$$

$$\alpha_u(r,s) - \alpha_v(r,s) + \beta_r(u,v) - \beta_s(u,v) = 0. \tag{7}$$

Equation (6) has $r < s < t$ and $u+2 \le r+s+t \le p+u+1$. Equation (7) has $r < s$, $u < v$, and $\max(0, R-p+4) \le r+s \le R+3$. Inadmissible source coefficients are zero. These equations exhaust the A and B products in (1).

For fixed u, read $f_u(r) = \alpha_u(0,r)$ when $r \ge u+2$ and set the remaining values to zero. Every admissible pair $0 < r < s$ occurs in the A equation with endpoint zero. It forces $\alpha_u(r,s) = f_u(s) - f_u(r)$. Pairs below the lower sum bound have both potentials zero. Pairs above the upper bound cannot occur in an A equation. Conversely these differences solve every A equation.

For a B equation, $r+s \le R+3 \le p+u+1$ since $v \le p-2$, so upper truncation cannot spoil the potential formula. Lower truncation has both relevant potentials zero. Equation (7) becomes $\beta_r - F(r) = \beta_s - F(s)$. If $R \le p-4$, the beta interval is $0, \dots, R+2$. The equation on $\{0, R+3\}$ forces $\beta_0 = -F(R+3)$, and the equations on $\{0, r\}$ give the displayed formula. If $R \ge p-3$, write $a = R-p+4 \ge 1$. Its beta interval is $a, \dots, p-1$, and $a \le u+2$. Both potentials vanish below a, so the equations with endpoint zero give $\beta_r = F(r)$. The remaining equations hold by subtraction. Reconstruction is unique and uses no division. The distinguished alpha coordinates therefore give an integral inverse and an identity coordinate minor.

For one unit potential, alpha has at most $p-1$ terms. Beta is supported only on second pairs containing its active second index. Usually such a pair contributes at most one term. At most one pair satisfies $R = r_0 - 3$ and contributes a constant interval of at most $p-1$ terms. Thus beta has at most $2p-4$ terms, and all coefficients have absolute value one.

Finally, every $6p$ face vanishes because $6p + L_0 \subset H_0$ and $6p + L_1 \subset H_1$. The negative $8p-4$ face survives only for first-low coefficient 3 or second-low coefficients $3p, 3p+1, 3p+2$. A unit potential gives at most one first case, on $r+s = p+u-1$. In the second case there are at most three ordinary partners, determined by $r_0 - u_0$, $r_0 - u_0 - 1$, $r_0 - u_0 - 2$, and at most three rows from the exceptional partner $r_0 - u_0 - 3$. This proves the full seven-row bound. $\square$

Remark 4.2. This is a complete basis theorem for one D kernel. It is not a normal form of coker M_p, nor a claim that the whole connecting presentation has seven rows. In particular, subsequent K relations may kill its images.

5 The shifted potential operator

For the annihilator, move the last high variable to $h = 8p - 2$. Let $f_u(r)$ be integral and vanish for $r \leq u$. Define $P(f)$ by the following two families, omitting zero terms and combining equal labels:

$$S[(L \setminus \{p-r, p-s, 3p+u\}) \cup \{6p, h\}; p+u-r-s], \quad \text{weight } (-1)^{p+r+s+u}(f_u(s) - f_u(r)), \quad r < s,\ u \leq r+s \leq p+u-1; \tag{8}$$

$$S[(L \setminus \{p-r, 3p+u, 3p+v\}) \cup \{6p, h\}; 3p+R-r], \quad \text{weight } (-1)^{p+r+u+v}\beta_r(u,v), \quad u < v,\ \max(0, R-p+2) \leq r \leq \min(p-1, R), \tag{9}$$

where $R = u + v$, $F = f_u - f_v$, and

$$\beta_r(u,v) = \begin{cases} F(r) - F(R+1), & R \leq p-2, \\ F(r), & R \geq p-1. \end{cases} \tag{10}$$

The coefficient offset is now the omitted-low sum minus $4p$, so all labels have total offset τ. The bounds put their coefficients in L_0 and L_1, respectively. Moving the last high variable does not change the exterior ordering sign. We assert a construction for these potentials, not a complete basis theorem for the shifted sector.

Lemma 5.1 ([D)**.** *Full shifted boundary] The D component of $M_pP(f)$ is zero. Every $6p$ face vanishes. The only other surviving faces are the negative h faces with first-low coefficient 1 or second-low coefficient $3p$.*

Proof. The A and B equations retain the signs (6) and (7), with bounds now

$$u \leq r+s+t \leq p+u-1, \qquad \max(0, R-p+2) \leq r+s \leq R+1.$$

In an A equation, a pair below its admissibility bound has both potentials zero, and no pair exceeds the upper bound. Its differences telescope. In a B equation, $r + s \leq R + 1 \leq p + u - 1$, so alpha again has no upper truncation. Its equation is $\beta_r - F(r) = \beta_s - F(s)$. For $R \leq p - 2$, formula (10) extends as zero at the absent endpoint $R + 1$. No endpoint in the equation can exceed it. For $R \geq p - 1$, the lower beta endpoint is $a = R - p + 2 \leq u$; both potentials vanish below a, so the second formula extends there by zero. This proves every D equation.

As before, the $6p$ products lie in H_0 or H_1. For $c \in L_0$, $h + c$ is in $\mathcal{B}$ only at $c = 1$, where it equals $8p - 1$. The remaining values lie in H_1. For $c = 3p + t \in L_1$, $h + c = 11p - 2 + t$ survives only at $t = 0$; the remaining values lie in H_3. The low exterior has even size $2p - 4$, so the last high face has negative sign. No other original boundary row occurs. □

Use the signed K row

$$e(r; u, v) = (-1)^{p+r+u+v} K[(L \setminus \{p-r, 3p+u, 3p+v\}) \cup \{6p\}; 11p - 2 + u + v - r], \tag{11}$$

only for admissible indices and degree-two offsets. Put $x_{uv} = e(u + v; u, v)$ when $u < v$ and $u + v \leq p - 1$. For orientation, let $y_{uv} = x_{uv}$ if $u < v$, $y_{uv} = -x_{vu}$ if $u > v$, and $y_{uu} = 0$. In particular, $x_{01} = e(1; 0, 1)$ is different from $e(2; 0, 1)$.

The C0 boundary in Lemma 5.1 lies on $r+s = p+u-1$. Its normalized coefficient is $-\alpha_u(r, s)$. The C2 boundary is diagonal: for $R \leq p-2$ its coefficient on x_{uv} is $F(R+1)-F(R)$; at $R = p-1$ it is $-F(p - 1)$. There is no such diagonal row for $R > p - 1$. This last endpoint cannot be replaced by an unrestricted difference formula.

6 Reflection intervals and the uniform annihilator

Lemma 6.1 ([D). *Signed interval triangle] Let i, a, b be distinct nonnegative second indices, with $a < b$ and $i + a + b = p - 2$. Let $T_i(a, b)$ have the single nonzero potential*

$$f_i(r) = 1 \quad \textit{for } i + a + 1 \le r \le i + b.$$

Then the exact original boundary is

$$M_p P(T_i(a, b)) = y_{ia} - y_{ib}. \tag{12}$$

Proof. The interval lies strictly inside $i < r < p - 1$ and its endpoints sum to $p + i - 1$. It is therefore invariant under $r \mapsto p + i - 1 - r$. Every possible C0 pair has this sum, so its two potential values agree, including pairs outside the interval. All C0 rows vanish. For C2 write $g(t) = f_i(i + t)$. Lemma 5.1 gives the oriented edge coefficient $g(v + 1) - g(v)$. The interval indicator jumps by $+1$ at $v = a$ and by -1 at $v = b$, and nowhere else. There is no top-endpoint contribution because the interval ends below $p - 1$. This proves (12) in the full target, not just after quotienting by K relations. □

For $j = 1, 2$, put $k = p - 2 - j$. Define the three-interval potential

$$\begin{aligned} (F_j)_0(r) &= 1 && \text{for } j + 1 \le r \le k, \\ (F_j)_j(r) &= -1 && \text{for } j + 1 \le r \le p - 2, \\ (F_j)_k(r) &= -1 && \text{for } k + 1 \le r \le p - 2, \end{aligned} \tag{13}$$

and set all other values to zero. The indices $0, j, k$ are distinct for every $p \ge 8$.

Proposition 6.2 ([D). *Twice-edge source] For $j = 1, 2$, $M_p P(F_j) = 2x_{0j}$.*

Proof. The definition is $F_j = T_0(j, k) - T_j(0, k) - T_k(0, j)$. Since $j < k$, Lemma 6.1 gives

$$M_p P(F_j) = (x_{0j} - x_{0k}) - (-x_{0j} - x_{jk}) - (-x_{0k} + x_{jk}) = 2x_{0j}.$$

All three interval sources have zero C0 boundary individually. The auxiliary edge satisfies $j + k = p - 2$, within the diagonal domain. □

Two original K-source corrections are required to recover the particular four-row target, rather than just the diagonal edges. Let δ_{0a} be the unit potential at $(u, r) = (0, a)$, and, for $a = 2, 3$, let

$$Q_a = K[(L \setminus \{p - a, 3p\}) \cup \{6p\}; 8p - 2 - a] \tag{14}$$

with *positive* unit coefficient.

Lemma 6.3 ([D). *Exact short corrections] Put $B = P(\delta_{03}) - Q_3$ and $D = P(\delta_{02}) + Q_2$. Then*

$$M_p B = e(3; 0, 1) + x_{02} + e(3; 0, 2), \tag{15}$$

$$M_p D = x_{01} + e(2; 0, 1). \tag{16}$$

Proof. Lemma 5.1 gives diagonal boundary $x_{0,a-1} - x_{0a}$ for $P(\delta_{0a})$. Its only C0 pair is $(a, p-1-a)$, with distinct increasing endpoints for $a = 2, 3$ and $p \ge 8$. The normalized alpha value is -1; its source sign is $(-1)^{2p-1} = -1$ and the high-face sign is negative. Thus the actual C0 coefficient is -1.

The only first-low face of Q_a surviving in $\mathcal{B}$ removes $a+1$, has coefficient offset $8p-1$, and sign $(-1)^a$. Indeed the omitted first-low variable $p-a$ is larger than $a+1$, so the latter has exterior position a. This is the same C0 row. The surviving second-low faces remove $3p+v$, $1 \le v \le a$. Their coefficient offset is $11p-2-a+v$, and their exterior sign is $(-1)^{p+v-2}$. In the signed convention (11), their coefficient on $e(a;0,v)$ is $(-1)^a$. Every other low product lies in H. The high face has coefficient offset $14p-2-a$; the two values $14p-4$ and $14p-5$ both lie in H_4 when $p \ge 8$. There is therefore no discarded high-free row or hidden filler.

Consequently the source $J_a = P(\delta_{0a}) + (-1)^a Q_a$ cancels C0 exactly. Its boundary is

$$M_p J_a = x_{0,a-1} + \sum_{v=1}^{a-1} e(a;0,v).$$

Specializing to $a=3$ and $a=2$ proves the lemma. ☐

Theorem 6.4 ([D). *Uniform original-map annihilation] For every integer $p \ge 8$, the explicitly defined integral source*

$$V_p = P(F_2 + 2F_1 - 4\delta_{03} - 4\delta_{02}) + 4Q_3 - 4Q_2 \tag{17}$$

satisfies $M_p V_p = 2\eta_p$. Hence both $[\eta_p]$ and $[b_p]$ have order dividing two in the full integral cokernel.

Proof. Changing (4) from raw to signed row labels gives

$$\eta_p = -2e(3;0,2) - x_{02} - 2e(2;0,1) - 2e(3;0,1).$$

Equations (15)–(16) therefore give the exact target-vector equality

$$\eta_p = x_{02} + 2x_{01} - 2M_p B - 2M_p D.$$

By Proposition 6.2, twice the right-hand side is

$$M_p\big(P(F_2) + 2P(F_1) - 4B - 4D\big) = M_p V_p.$$

Every term is an admissible original source and every boundary row has been included. Proposition 3.1 also gives the explicit identity $M_p(2s_p + 2q_p - V_p) = 2b_p$. ☐

Corollary 6.5 ([D). *Base change with two invertible] Let R be any commutative coefficient ring in which 2 is a unit. In the cokernel of $M_p \otimes_{\mathbb{Z}} R$, the images of η_p and b_p vanish, with explicit sources $V_p/2$ and $s_p + q_p - V_p/2$, respectively.*

Proof. Tensor the two original source identities with R and multiply by the unit 2^{-1}. This argument concerns only the two tracked classes. ☐

7 A relative parity functional and exact order two

All coefficients in this section lie in $\mathbb{F}_2$. Write the full map as

$$M_p(s,t) = (d_D s,\ d_K^S s + d_K^K t).$$

A functional on the K target need not annihilate every S column individually. It suffices to annihilate d_K^K and $d_K^S(\ker d_D)$ while taking value one on η_p. This relative formulation is essential: no functional on a truncated matrix is substituted for a statement about the full map.

7.1 The reflection template and its twelve-row specialization

Let T be any unordered triple of distinct indices in $[0, p-2]$ whose sum is $p-2$. Define

$$z_{uv}^T = \begin{cases} 1, & \{u, v, p-2-u-v\} \text{ is the multiset } T, \\ 0, & \text{otherwise.} \end{cases}$$

Invalid indices and diagonals have value zero. Write $z = z^T$ temporarily. It is symmetric and has the reflection identity

$$z_{uv} = z_{vu} = z_{u,p-2-u-v}. \tag{18}$$

Assign the row $e(r; u, v)$ the value z_{uv} when $r = u+v$ or $u+v+1$, and zero otherwise. On a C0 row

$$K[(L \setminus \{p-r, p-s, 3p+u\}) \cup \{6p\}; 8p-1], \qquad r < s, \quad r+s = p+u-1,$$

assign value

$$d_u(r) = z_{u,r-u} + z_{u,r-u-1}. \tag{19}$$

Set every other original K row to zero. This defines λ_T.

The C0 definition does not depend on which endpoint is used. If $n = p-1-u$ and $t = r-u$, then $s-u = n-t$, and (18) exchanges the two summands of $d_u(r)$ and $d_u(s)$. At a reflected fixed point $r = s$, the summands coincide and $d_u(r) = 0$. Thus no repeated exterior variable is introduced. Possible cancellations in (19) are retained. More precisely, write $T = \{i, j, k\}$ with $i < j < k$. For each $u \in T$, let $v < w$ be its other two entries. The C0 pairs are

$$(r, s; u) = (u+v, u+w+1; u), \qquad (u+v+1, u+w; u).$$

The second pair is omitted if $w = v+1$: it is a fixed point and its two values cancel modulo two. All endpoints lie in $[0, p-1]$, and distinct u give distinct omitted second-low variables. There are always six distinct C2 rows, so the exact support size is

$$|\operatorname{supp} \lambda_T| = 12 - \mathbf{1}_{j=i+1} - \mathbf{1}_{k=j+1}. \tag{20}$$

It ranges from ten to twelve; for example $T = \{1, 2, 3\}$ at $p = 8$ has ten rows. No numerical support count is assumed in this argument.

For the tracked class choose $T = \{0, 2, k\}$, where $k = p-4$, and write $\lambda_K = \lambda_T$. Its six C2 rows are

$$e(S; u, v),\ e(S+1; u, v), \qquad (u, v) = (0, 2), (0, k), (2, k), \quad S = u+v.$$

Its six C0 rows have omitted endpoints

$$\begin{gathered}(r, s; u) = (2, p-3; 0), (3, p-4; 0), (2, p-1; 2), \\ (3, p-2; 2), (p-4, p-1; k), (p-3, p-2; k).\end{gathered} \tag{21}$$

All twelve rows are distinct and admissible for $p \geq 8$. The C0 omitted-low sum is $4p+1$, which verifies its target grading; C2 grading follows from (11). The two coefficient types are distinct. Modulo two, (4) is just $e(2; 0, 2)$, so

$$\lambda_K(\eta_p) = z_{02} = 1. \tag{22}$$

7.2 All original K sources

Lemma 7.1 ([D). *Complete K-source annihilation] Every reflection template λ_T annihilates d_K^K.*

Proof. A source with no exterior high variable cannot reach a supporting row. With at least two exterior highs, a low face retains too many highs, while a high face has coefficient at least $12p$, larger than every supported coefficient. With one high different from $6p$, a low face retains the wrong high and a high face has none. It remains only to consider exterior high set $\{6p\}$.

Its low exterior omits two elements of L. If their sum is m, grading forces coefficient $4p-2+m$. Two omitted first-low endpoints $r < s$ would give $6p-2-r-s < 6p$, not an original K coefficient. One omitted first-low and one second-low variable gives

$$X(r,u) = K[(L \setminus \{p-r, 3p+u\}) \cup \{6p\}; 8p-2+u-r].$$

The only excluded high value in its possible coefficient range is $8p-1$, so this column exists precisely when $r \geq u$ or $r \leq u-2$. If $r \leq u-2$, neither supported C2 endpoint is possible and the other C0 endpoint would exceed $p-1$. If $r \geq u$, its possible C0 face has reflected endpoint $s = p+u-1-r$ and value $d_u(r)$. At $s = r$ the face is absent but its assigned value is zero. Its C2 faces have second indices $r-u$ and $r-u-1$, with total value again $d_u(r)$; invalid or already omitted indices have value zero. The two contributions cancel, and the high face is invisible.

Finally, two omitted second-low indices $u < v$, of sum S, force coefficient $10p-2+S$. This is in H precisely for $S = 2$ or $S \geq p+1$. In the first case both supported first endpoints 2 and 3 are present and have the same value z_{uv}. In the second neither S nor $S+1$ is an available endpoint. Other faces are invisible. This exhausts every original K source. □

7.3 Reachability and complete D kernels

An S source can reach the functional only by deleting a high variable from an exterior high set $\{6p, h\}$. If b is a supported target coefficient and $c \in L$ the source coefficient, then $h = b - c \in H$. The complete intersection calculation is

b	$c \in L_0$	$c \in L_1$
$8p-1$	$[7p-1, 8p-2]$	none
$11p-2$	$\{10p-2, 10p\}$	$[7p, 8p-2]$
$11p-3$	$\{10p-3, 10p-2, 10p\}$	$[7p-1, 8p-3]$

Thus only

$$h \in [7p-1, 8p-2] \cup \{10p-3, 10p-2, 10p\} \tag{23}$$

can matter. In particular $10p-3$ is present: first-low coefficient p reaches $11p-3$. The D differential preserves the complete exterior-high set. Its kernel therefore splits into kernels of the individual high-set sectors, and all sectors outside (23) have zero pairing term by term.

Consider first $h = 8p-d$, with $2 \leq d \leq p+1$. A source omits three low variables of sum m, with coefficient $m - 4p + d - 2$. Three omitted first-low variables make this at most $d - p - 5 < 1$; three omitted second-low variables make it at least $5p + d + 1 > 4p - 2$. The two exhaustive source families and their coefficients are

$$\begin{aligned} &\alpha_u(r,s): \quad p+d-2+u-r-s, && u+d-2 \leq r+s \leq p+u+d-3, \quad r < s; \\ &\beta_r(u,v): \quad 3p+d-2+S-r, && \max(0, S+d-p) \leq r \leq \min(p-1, S+d-2), \quad u < v. \end{aligned}$$

They have the same omitted-variable patterns as the alpha and beta sources of Section 4. Alpha coefficients can only lie in L_0, beta coefficients only in L_1.

Proposition 7.2 ([D]). *Complete generalized potential reconstruction] Every D cycle in this sector is uniquely parametrized over $\mathbb{F}_2$ by potentials satisfying*

$$f_u(0) = 0, \qquad f_u(r) = 0 \text{ for } r < u + d - 2.$$

Its coefficients are

$$\alpha_u(r, s) = f_u(r) + f_u(s), \tag{24}$$

$$\beta_r(u, v) = \begin{cases} F(r) + F(S + d - 1), & S \leq p - d, \\ F(r), & S \geq p - d + 1, \end{cases} \qquad F = f_u + f_v. \tag{25}$$

In particular, for $d = 2$ the coordinates $f_u(u)$ are free for every $u \geq 1$. Only $f_0(0)$ is excluded by the endpoint-zero convention.

Proof. The complete A equations are

$$\alpha_u(r, s) + \alpha_u(r, t) + \alpha_u(s, t) = 0, \quad u + d - 2 \leq r + s + t \leq p + u + d - 3.$$

Set $f_u(r) = \alpha_u(0, r)$ for $r > 0$ with $r \geq u + d - 2$, and zero otherwise. The pair's upper bound is at least $p - 1$. Every other admissible pair occurs in its A equation with endpoint zero, which forces (24). Conversely these potentials solve all A equations: a pair below the lower bound has both potentials zero, and a pair above the upper bound cannot occur in an admissible triple.

The complete B equations are

$$\alpha_u(r, s) + \alpha_v(r, s) + \beta_r(u, v) + \beta_s(u, v) = 0, \quad \max(0, S + d - p) \leq r + s \leq S + d - 1.$$

Since $S+d-1 \leq p+u+d-3$, both alpha terms can be replaced by their potential sums, including lower truncations. The equation becomes $\beta_r + F(r) = \beta_s + F(s)$.

If $S \leq p-d$, the beta interval is $0, \ldots, b$ with $b = S+d-2 \leq p-2$. The equation on $\{0, b+1\}$ forces $\beta_0 = F(b+1)$; the other endpoint-zero equations force the first formula in (25). It extends as zero at $b+1$ and solves every B equation. If $S \geq p - d + 1$, put $a = S + d - p \geq 1$. The beta interval is $a, \ldots, p-1$, possibly empty, and $a \leq u + d - 2$. Both potentials vanish below a, so the endpoint-zero equations force $\beta_r = F(r)$; this extends by zero below a. If $a > p - 1$, both potentials vanish everywhere and there is no beta source. Thus all truncations, empty intervals, and converses are checked. The reconstruction reads original coordinates, not a rank pattern. ☐

The additional free coordinates at $d = 2$ cannot be suppressed here. Section 5 used a restricted integral construction with $f_u(r) = 0$ for $r \leq u$; nonvanishing requires the complete kernel, including its previously unused diagonal coordinates. The first-low endpoint zero used in the reconstruction is available even when the smallest second-low index in T is positive. It places no restriction on the triangle.

7.4 Pairing the complete kernels and the remaining sectors

For a supported C0 face in a generalized potential cycle, the source coefficient is $d - 1$ and $r + s = p + u - 1$. Reflection makes its value $(f_u(r) + f_u(s))d_u(r)$. Fixed points contribute zero. Every nonzero potential has $r \geq u$, so its reflected endpoint is also available. Summing all distinct pairs gives

$$\sum_{u,r} f_u(r) d_u(r) = \sum_{u<v} z_{uv}(F(u + v) + F(u + v + 1)). \tag{26}$$

On the support of a triple template, $1 \le S = u + v \le p - 2$, so both potential indices in this sum are available.

For C2, only $r = S$ and $r = S + 1$ can pair. Their source coefficients are $3p + d - 2$ and $3p + d - 3$. If $d = 2$, only $r = S$ is admissible; (25) gives $F(S) + F(S + 1)$. If $3 \le d \le p$, both endpoints are admissible, and their common beta constant, when present, cancels. If $d = p + 1$, only $r = S + 1$ is admissible. The sole possible nonzero potential is $f_0(p - 1)$, so $F(S) = 0$ and the same expression remains. Thus C2 pairing equals (26). It cancels C0 over $\mathbb{F}_2$. Every other K face has functional value zero; it need not itself vanish.

It remains to treat $h = 10p-3, 10p-2, 10p$. Sources omitting two first-low and one second-low variable cannot occur: their omitted sum is at most $6p - 3$, giving coefficient at most -2 even for $h = 10p - 3$. Three omitted first-low variables also cannot occur. Sources omitting three second-low variables, if admissible, have second-low coefficients and do not contribute to A rows. Therefore the only source family with first-low coefficient omits one first-low and two second-low variables. For fixed $u < v$ and $S = u + v$, write its coefficients as a_r. Its coefficient offsets are respectively

$$p + 1 + S - r, \qquad p + S - r, \qquad p - 2 + S - r.$$

The lower endpoint bounds are $r \ge S + 1$, $r \ge S$, and $r \ge \max(0, S - 2)$.

For each existing $r > 0$, removal of the first-low variable p gives an A equation on endpoints $\{0, r\}$. Its only possible other contributor is a_0. In the first two high sectors a_0 is absent, so these equations force every a_r to zero. For $h = 10p$ the same argument applies when $S \ge 3$. For the remaining possible sums $S = 1, 2$, use the A row on endpoints $\{S, S + 1\}$. Its coefficient offset is $2p - 3 - S > p$ and its only contributors are $a_S + a_{S+1}$, which therefore vanishes. These are exactly the two supported C2 contributions. This also covers triple templates containing $\{0, 1\}$; the selected twelve-row template itself has sums $2, p - 4, p - 2$. There is no C0 contribution in these three high sectors. Other families may contribute to B equations but cannot change the A constraints used here.

We have proved the following relative statement on the complete map.

Proposition 7.3 ([D]**.** *Relative reflection functional]* *For every distinct triple template T above,*

$$\lambda_T d_K^K = 0, \qquad \lambda_T d_K^S(s) = 0 \quad \textit{whenever } d_D s = 0.$$

This statement applies to every triangle, including those with positive smallest entry; Section 8 uses it to establish independence.

Theorem 7.4 ([D]**.** *Exact order of the tracked class]* *For every $p \ge 8$, η_p is nonzero in the cokernel of $M_p \otimes \mathbb{F}_2$. Its class in $\operatorname{coker}_{\mathbb{Z}} M_p$ has exact order two. The same is true of the class of b_p.*

Proof. Suppose $\eta_p = M_p(s, t)$ over $\mathbb{F}_2$. Its zero D component implies $d_D s = 0$. Lemma 7.1 and Proposition 7.3, for $T = \{0, 2, p - 4\}$, would give

$$1 = \lambda_K(\eta_p) = \lambda_K(d_K^S s + d_K^K t) = 0,$$

contradicting (22). An integral source for η_p would reduce to a source modulo two, so its integral class is also nonzero. Theorem 6.4 proves twice the class is zero. Its order is therefore exactly two. Proposition 3.1 transfers the conclusion to $[b_p] = -[\eta_p]$. ∎

8 The quadratic triangle family and its splitting

Return to integral coefficients for source identities, and put $n = p - 2$. Let

$$\mathcal{T}_p = \{(i,j,k) : 0 \le i < j < k,\ i+j+k = n\}, \qquad \mathcal{C}_p = \operatorname{coker}_{\mathbb{Z}} M_p.$$

For $T = (i,j,k)$ select the original signed row $x_T = x_{ij}$ from (11). Its first endpoint $i+j$ is at most $p-2$, and its coefficient offset is $11p-2$, so it is an admissible original target. All subsequent relation statements take place in $\mathcal{C}_p$, not in a projected or isolated subpresentation.

Proposition 8.1 ([D). *Full original triangle sources] For every $T = (i,j,k) \in \mathcal{T}_p$, define*

$$F_T = T_i(j,k) - T_j(i,k) - T_k(i,j), \qquad W_T = P(F_T). \tag{27}$$

Here each $T_u(a,b)$ is the integral interval potential of Lemma 6.1, and P retains exactly the source labels and signs in (8)–(10). Then $M_p W_T = 2x_T$ in the full original target.

Proof. For distinct u,a,b with $a<b$ and $u+a+b=n$, the interval $[u+a+1, u+b]$ is nonempty, lies strictly above u, and ends at $n-a \le p-2$. Its endpoints sum to $p+u-1$. Thus every interval in (27) satisfies all hypotheses of Lemma 6.1, regardless of whether $i=0$. The complete D boundary cancels by the signed potential equations; every $6p$ face vanishes; reflection kills C0; and the two interval jumps give precisely its oriented C2 edges. Consequently

$$\begin{aligned} M_p W_T &= (x_{ij} - x_{ik}) - (-x_{ij} - x_{jk}) - (-x_{ik} + x_{jk}) \\ &= 2x_{ij}. \end{aligned}$$

All coefficients here are integers. No unsupported boundary row has been dropped, and no subsequent K relation is needed for the equality. □

Theorem 8.2 ([D). *Independent triangle classes and exact count] For every $p \ge 8$, the relation lattice of the classes $([x_T])_{T\in\mathcal{T}_p}$ in $\mathcal{C}_p$ is exactly $2\mathbb{Z}^{\mathcal{T}_p}$. In particular there is an injection*

$$\iota : (\mathbb{Z}/2)^{\mathcal{T}_p} \hookrightarrow \mathcal{C}_p, \qquad \mathbf{e}_T \longmapsto [x_T], \qquad |\mathcal{T}_p| = \left\lfloor \frac{(p-2)^2+3}{12} \right\rfloor. \tag{28}$$

Proof. An unordered edge $\{a,b\}$ belongs to at most one triple of sum n: its third entry must be $n-a-b$. Thus different triangles have disjoint unordered edge sets. By the C2 definition of the relative functionals,

$$\lambda_T(x_U) = \delta_{TU} \quad \text{over } \mathbb{F}_2. \tag{29}$$

Suppose that $\sum_U m_U x_U = M_p(s,t)$ integrally. Reducing the full source equation modulo two, its zero D component gives $d_D s = 0$. Proposition 7.3 therefore forces every λ_T to evaluate to zero on the right-hand K component. Equation (29) gives $m_T \equiv 0 \pmod 2$ for every T. Conversely, every even coefficient vector is a relation by Proposition 8.1. This proves the exact relation lattice and the injection, not merely nonzero target vectors.

To count the triples, let $q(n)$ count strictly increasing nonnegative triples of sum n. For $n \ge 6$, triples with smallest entry at least two correspond bijectively to those of sum $n-6$, by subtracting two from every entry. Those with smallest entry zero number $\lfloor (n-1)/2 \rfloor$, and those with smallest entry one number $\lfloor (n-2)/2 \rfloor - 1$. Hence

$$q(n) - q(n-6) = n - 3 \qquad (n \ge 6). \tag{30}$$

The initial counts for $n = 0, \ldots, 5$ are $0,0,0,1,1,2$. The function $\lfloor (n^2+3)/12 \rfloor$ has those same values and the same recurrence, since the two unrounded arguments differ by the integer $n-3$. Induction in each residue class modulo six proves the formula for all $n \ge 0$, hence for $n = p-2$. □

Corollary 8.3 ([D). *A direct summand of integral two-torsion] The injection* (28) *splits as a homomorphism of abelian groups. Thus, for some unspecified abelian group $\mathcal{C}'_p$,*

$$\mathcal{C}_p \cong (\mathbb{Z}/2)^{\lfloor((p-2)^2+3)/12\rfloor} \oplus \mathcal{C}'_p.$$

This is an existence statement for a retraction, not an explicit uniform formula for its values on all D rows or a complete cokernel computation.

Proof. Work first over $\mathbb{F}_2$. For each T, define a functional on the subspace $\operatorname{im} d_D$ of the full D target by

$$\mu_T(d_D s) = \lambda_T(d_K^S s).$$

It is well-defined: two choices of s differ by a complete D cycle, whose connecting image is annihilated by Proposition 7.3. Extend μ_T linearly to the entire D target, by completing a basis of $\operatorname{im} d_D$. The target is finite-dimensional for each fixed p, so such an extension exists. The functional (μ_T, λ_T) annihilates the full map: on an S source its two values are equal and cancel over $\mathbb{F}_2$, and on a K source the value is zero by Lemma 7.1.

Reduction modulo two followed by these extended functionals therefore defines a group homomorphism $\rho : \mathcal{C}_p \longrightarrow (\mathbb{Z}/2)^{\mathcal{T}_p}$. Every selected target x_U has zero D component, so (29) gives $\rho\iota = \mathrm{id}$. Thus $\mathcal{C}_p = \operatorname{im} \iota \oplus \ker \rho$. The extension is noncanonical; no global D-row certificate is asserted. $\square$

Proposition 8.4 ([D). *Exact transfer of the tracked class] The original source*

$$C = P(F_1) - 2B - 2D = P(F_1 - 2\delta_{03} - 2\delta_{02}) + 2Q_3 - 2Q_2 \tag{31}$$

satisfies $M_p C = \eta_p - x_{02}$. Hence $[\eta_p] = [x_{02}]$ and $[b_p] = -[x_{02}]$ in $\mathcal{C}_p$, where x_{02} is chosen for the triangle $(0, 2, p-4)$*.*

Proof. The target identity in the proof of Theorem 6.4 is $\eta_p = x_{02} + 2x_{01} - 2M_pB - 2M_pD$. Apply $M_pP(F_1) = 2x_{01}$ and then Proposition 3.1. This proves a quotient equality with an explicit source for its difference, not equality of the original target vectors η_p and x_{02}. $\square$

In particular there are at least $3, 4, 5, 7$ independent integral two-torsion classes at $p = 8, 9, 10, 11$, respectively, and their number is quadratically unbounded. If two is invertible in a coefficient ring, each selected class vanishes with source $W_T/2$; this gives no vanishing claim for the complementary quotient. Agreement of these lower bounds with earlier finite isolated ranks does not identify that isolated object with $\mathcal{C}_p$, nor does it establish an upper bound.

9 Local obstruction and exact validation

9.1 A failed locality prediction

Before the interval construction, EXP-058 explored original source columns incident to the four endpoint rows, retaining all their boundary rows. Each row-column-row round gives the next radius. At $p = 8$, the complete declared radius-two span excludes $2\eta_8$ even over $\mathbb{Q}$ [MV]. The decisive certificates are integer row functionals:

Radius	Source columns	Boundary rows	Incidences	Dual support
1	47	240	306	14
2	330	1803	2669	40

For each selected matrix M_{loc}, the stored functional satisfies $\lambda M_{\mathrm{loc}} = 0$ and $\lambda(2\eta_8) = 4$. The radius-two functional has 35 D rows and five K rows, all nonzero coefficients equal to 1 or -1. A coefficient-first reconstruction of the complete neighborhoods, a separate original differential, and a dual mutation check verify the certificates independently. The experiment stopped at $p = 8$; its planned $p = 9, 10$ cases were not run.

If a full source z satisfies $M_8 z = 2\eta_8$, then some column outside the selected radius-two set must have nonzero coefficient in z and nonzero pairing with λ. Otherwise applying λ gives $0 = 4$. This elementary escaping-column conclusion is compatible with Theorem 6.4: the explicit source is not restricted to that neighborhood. The local dual is not a functional on the full cokernel, and its even target pairing is not a characteristic-two nonvanishing certificate.

9.2 Campaigns and adversarial controls

For Theorem 4.1, EXP-059 checks every unit potential at $p = 8, \ldots, 16$, giving 525 chains, and four frozen endpoint pairs at each $p = 17, \ldots, 100$, giving 336 more [MV]. All 861 full original boundaries agree with the seven-row formula. The producer uses both left-to-right and independently encoded right-to-left differentials. A separate coefficient-first source reconstruction and bitset differential audit all 861 cases; the latter also crosschecks the literal differential on the 525 small cases. Source support ranges from 9 to 197 and boundary support from 3 to 7 in this declared campaign. The integral completeness claim, however, follows from the coordinate reconstruction proof.

For Theorem 6.4, EXP-060 checks the fixed parameters

$$8, 9, \ldots, 20, 25, 32, 50, 64, 100.$$

At all 18 values, two original differential implementations verify the potential components, both short corrections, and $M_p V_p = 2\eta_p$ [MV]. The independent auditor reconstructs all source labels and coefficients, verifies the full boundary by a separately encoded bitset method, and performs 25 literal differential crosschecks on five sources at each $p = 8, \ldots, 12$. The complete boundary has four K rows and no D rows. Observed source support is 110 at $p = 8$ and 3054 at $p = 100$, with coefficient height five at each checked value. These are finite support observations, not a separately proved uniform counting formula.

Wrong signs and single-coefficient changes are rejected. Two pre-declaration paper errors are retained explicitly. Omitting the δ_{02}/Q_2 correction gives $V_{\mathrm{omit}} = V_p + 4D$ and

$$M_p V_{\mathrm{omit}} - 2\eta_p = 4(x_{01} + e(2; 0, 1)) \neq 0.$$

The earliest proposal also had the opposite F_1 sign, giving $V_{\mathrm{early}} = V_{\mathrm{omit}} - 4P(F_1)$ and

$$M_p V_{\mathrm{early}} - 2\eta_p = 4(e(2; 0, 1) - x_{01}) \neq 0.$$

The rows are distinct, so both failures hold uniformly. They were corrected on paper before the final hypothesis was committed, not hidden after an experimental failure.

Each producer and audit completed within its declared 60-second, one-CPU-process and 1-GiB private-memory cap. There is no new HNF, Smith-form or full ambient-basis computation in these campaigns. No old labelled HNF source, including the original $p = 11$ source holdout, is an input. Checking the new formula at $p = 11$ is not described as a new untouched holdout. The dedicated suites contain 12 tests for the local obstruction and 20 each for the potential basis and annihilator; tests write only temporary outputs.

9.3 Full-sector parity verification

EXP-061 checks the twelve-row functional and 2,123 complete-sector unit potential chains at $p = 8, \ldots, 16, 25, 32, 50, 64, 100$ [MV]. Every basis element is checked at $p \leq 12$; larger parameters use frozen first, middle and last basis indices, together with the newly free $d = 2$ diagonal coordinate $(u, r) = (1, 1)$. All chains have zero complete D boundary and equal C0 and C2 pairings. The original K inverse-incidence enumeration at $p = 8, \ldots, 12$ finds 13 incident K columns at each parameter and annihilates every one. The reachable high-sector lists include $10p - 3$. The producer's 21 dedicated tests include the free diagonal cases $(u, r) = (2, 2)$ and $(4, 4)$ at $p = 8$, where the two individual pairings are both one and cancel.

The independent audit does not use the producer's potential formulas. It enumerates every original S source in every reachable high sector at $p = 8, \ldots, 12$, constructs all original D rows over $\mathbb{F}_2$, and certifies that the K pairing lies in the row span of that complete D matrix. Equivalently, the pairing vanishes on its entire kernel. The totals are 65 sectors, 23,695 original S sources, 108,261 D rows and 231,986 incidences. The saved original-D-row dual certificates contain 1,070 terms in total. Complete kernel dimensions agree with the potential reconstruction, including the additional $d = 2$ freedom.

All 20 independent adversarial controls pass. They include removal of the odd eta-pairing row, mutation of a support index, an explicit diagnostic for the otherwise omitted $h = 10p - 3$ sector, and a functional that passes a proper local column subset but fails an added original column. The producer and auditor each finish within their separately declared 120-second, one-process, 1-GiB caps. These finite checks challenge the full-sector argument, not just a local span; the proof of Theorem 7.4 remains the source classification and relative-functional contradiction above.

9.4 All-triangle identities and independent sector certificates

EXP-062 verifies all 49 triangle sources at $p = 8, \ldots, 14$ and the first, middle and last lexicographic triangle at each of $p = 16, 20, 25, 32, 50, 64, 100$, for 70 full integral identities [MV]. The producer also checks seven complete small-parameter pairing matrices, 93 count identities at $p = 8, \ldots, 100$, and the five original $\eta_p - x_{02}$ transfers at $p = 8, \ldots, 12$. Both original differential encodings agree on every signed source and transfer boundary. The source selection was frozen before computation; the count theorem at $p = 100$ does not mean that every source at that parameter was run.

The independent auditor reconstructs all 70 signed sources, all five transfers, the pairings, counts and mutation controls without importing producer mathematics. For every one of the 27 triangles at $p = 8, \ldots, 12$, it additionally enumerates all original S columns in every reachable high sector and gives exact $\mathbb{F}_2$ row-span certificates on the complete D matrix. The totals are

Complete-audit quantity	Count
Distinct (p, h) sectors	65
Distinct original S sources	23,695
Triangle-sector certificates	364
S-source instances across certificates	151,319
D-row instances across certificates	713,511
D incidences across certificates	1,528,426
Labelled original D-row dual terms	5,550

Instance counts repeat a physical sector for different triangle functionals; they are not counts of distinct matrices or columns. All original D rows are retained in each certificate. This tests complete-kernel annihilation, not merely the producer's constructed potential chains.

The ten-row adjacency case $T = (1, 2, 3)$ at $p = 8$ is included. Removing a functional row is detected by an explicit original K column. Changed source signs or coefficients have nonzero full unit-column residuals; permanent tests also multiply the actually altered sources. Duplicate and mirrored edge choices give singular pairing matrices. All 39 combined tests pass, comprising 24 producer and 15 independent-audit tests, and the independent audit replays byte-for-byte to temporary outputs.

Observed source support ranges from 38 to 13,594, with coefficient height two throughout this campaign. These are finite observations, whereas (20) proves the ten-to-twelve-row functional bound uniformly. Producer and auditor each finish within their own declared 120-second, one-process, 1-GiB caps. No dense full ambient matrix, integral HNF/SNF or old labelled $p = 11$ HNF source is used. Finite certification and the uniform proof remain separate layers of evidence.

10 Remaining obligations

The uniform twice-source, nonvanishing and independent-family obligations are closed: Theorem 8.2 and Corollary 8.3 give the quadratically growing direct summand for every $p \geq 8$. The tracked endpoint class is explicitly identified with one selected triangle class. What remains [C] is the complementary group: the complete cokernel, a matching upper bound, its free rank and any additional torsion are not determined. The functionals are not asserted to exhaust the full dual space. In particular, the theorem does not identify the full presentation with an earlier isolated component or relative-completion quotient, and it does not prove a lower-strand recurrence.

The mathematical premise imported from the earlier family is the integral coefficient presentation itself. Independent encodings attack implementation, sign and boundary errors but do not rederive that family-to-presentation identification. Within the explicit presentation of Section 2, the proofs above are self-contained. The new potential basis, signed interval triangle and relative parity proof provide explicit sources and detectors for further reductions. The parity argument exhausts the sectors visible to each functional; it does not assert that those sectors exhaust the whole map. The splitting is a proved existence result, but computing a uniform global D-row retraction remains a separate constructive task.

A Evidence map and source persistence

All evidence is in the public CAOS_RESEARCH repository, under the Huneke–Wiegand experiment tree. The following links lead to the primary declarations, proofs, verdicts, source code and exact artifacts. Mathematical claims are owned by those primary records, not by a resume or summary page.

Claim in this paper	Primary record
Original presentation and product rules	EXP-036, EXP-054
Proposition 3.1 and four-row target	EXP-056, EXP-057
Local radius-two exclusion and duals	EXP-058
Theorem 4.1, 861-chain audit	EXP-059
Lemmas 5.1–6.3, Theorem 6.4, Corollary 6.5	EXP-060
Complete relative parity proof and Theorem 7.4	EXP-061
Theorem 8.2, Corollary 8.3, Proposition 8.4 and complete triangle audits	EXP-062

The full EXP-060 source archive retains every original exterior label and integer coefficient for all 18 declared parameters. The EXP-062 archive similarly retains all 70 tested triangle sources and five transfer sources. Both deterministic gzip encodings have zero timestamp and empty embedded filename. Their compact manifests store compressed and raw SHA-256 hashes, byte counts, parameter counts, and per-source hashes. The auditors decompress the archives and independently reconstruct all labels; no mathematical information is discarded by compression.

Artifact	Raw bytes	Stored bytes	SHA-256 prefix
EXP-060 source archive	4,508,227	80,250	`85f8ea0800f0e8b6...`
EXP-060 result manifest	462,616	462,616	`f57fa704d045a8a8...`
EXP-060 independent audit	exact JSON certificate		`3bcf3cabe5b0330f...`
EXP-062 source archive	17,253,771	252,812	`8475ffb305c1567e...`
EXP-062 independent audit	exact JSON certificate		`854405294fab1d88...`

Archive prefixes in this table identify the compressed bytes. The EXP-060 raw digest starts with `d70c6ce4d7b331ef`; the EXP-062 raw digest starts with `af39d545d2f664de`. Prefixes are navigation aids, not sufficient integrity checks. Reproduction must verify the full 64-digit hashes in the manifest and audit certificate. The accompanying permanent tests also verify deterministic gzip replay, all stored source hashes, failure checkpoint preservation, and the original b_p source transfer at $p = 8$.

References

[1] F. Santibañez-Leal, *Frobenius Minimality, Minimum-Layer Uniqueness, and an Infinite Family of Huneke–Wiegand Counterexamples*, version 0.23, CAOS Research preprint, 2026. doi:10.5281/zenodo.22181972.

[2] W. Bruns and J. Herzog, *Semigroup rings and simplicial complexes*, Journal of Pure and Applied Algebra **122** (1997), 185–208. Author-hosted published paper.